

EXACT ANSWERS ONLY. Show your work for full credit!!!

6-1 Learning Goal: I can find and simplify the nth root.

Express in simplest form.

1. $\sqrt{40}$ $\begin{array}{c} \wedge \\ 2 \cdot 20 \\ \wedge \\ 2 \cdot 10 \\ \wedge \\ 2 \cdot 5 \\ \uparrow \\ = \sqrt{2 \cdot 2 \cdot 2 \cdot 5} \\ = \boxed{2\sqrt{10}} \end{array}$	2. $\sqrt{150}$ $\begin{array}{c} \wedge \\ 2 \cdot 75 \\ \wedge \\ 3 \cdot 25 \\ \wedge \\ 5 \cdot 5 \\ \uparrow \\ = \sqrt{2 \cdot 3 \cdot 5 \cdot 5} \\ = \boxed{5\sqrt{6}} \end{array}$	3. $\sqrt[3]{128}$ $\begin{array}{c} \wedge \\ 2 \cdot 64 \\ \wedge \\ 2 \cdot 32 \\ \wedge \\ 2 \cdot 16 \\ \wedge \\ 2 \cdot 8 \\ \wedge \\ 2 \cdot 4 \\ \wedge \\ 2 \cdot 2 \\ \uparrow \\ = \sqrt[3]{2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2} \\ = \boxed{4\sqrt[3]{2}} \end{array}$
4. $\sqrt[3]{96}$ $\begin{array}{c} \wedge \\ 2 \cdot 48 \\ \wedge \\ 2 \cdot 24 \\ \wedge \\ 2 \cdot 12 \\ \wedge \\ 2 \cdot 6 \\ \wedge \\ 2 \cdot 3 \\ \uparrow \\ = \sqrt[3]{2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 3} \\ = \boxed{2\sqrt[3]{12}} \end{array}$	5. $\sqrt{81a^8b^5}$ $\begin{array}{c} \wedge \quad \wedge \quad \wedge \quad \wedge \\ 3 \cdot 3 \cdot 3 \cdot 3 \cdot a^8 \cdot b^4 \cdot b^1 \\ \uparrow \\ = \sqrt{3 \cdot 3 \cdot 3 \cdot 3 \cdot a^8 \cdot b^4 \cdot b^1} \\ = \boxed{9a^4b^2\sqrt{b}} \end{array}$	6. $\sqrt[3]{512x^{10}}$ $\begin{array}{c} \wedge \\ 2 \cdot 256 \\ \wedge \\ 2 \cdot 128 \\ \wedge \\ 2 \cdot 64 \\ \wedge \\ 2 \cdot 32 \\ \wedge \\ 2 \cdot 16 \\ \wedge \\ 2 \cdot 8 \\ \wedge \\ 2 \cdot 4 \\ \wedge \\ 2 \cdot 2 \\ \uparrow \\ = \sqrt[3]{2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2} \\ = \boxed{8x^3\sqrt[3]{x}} \end{array}$
7. $4\sqrt[3]{52a^7}$ $\begin{array}{c} \wedge \\ 2 \cdot 26 \\ \wedge \\ 2 \cdot 13 \\ \uparrow \\ = 4\sqrt[3]{2 \cdot 2 \cdot 13 \cdot a^6 \cdot a^1} \\ = \boxed{4a^2\sqrt[3]{52a}} \end{array}$	8. $3\sqrt{128a^6}$ $\begin{array}{c} \wedge \\ 2 \cdot 64 \\ \wedge \\ 2 \cdot 32 \\ \wedge \\ 2 \cdot 16 \\ \wedge \\ 2 \cdot 8 \\ \wedge \\ 2 \cdot 4 \\ \wedge \\ 2 \cdot 2 \\ \uparrow \\ = 3\sqrt{2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot a^6} \\ = \boxed{24a^3\sqrt{2}} \end{array}$	9. $y^2\sqrt{121x^8y^5}$ $\begin{array}{c} \wedge \\ 11 \cdot 11 \\ \uparrow \\ y^2 \sqrt{11 \cdot 11 \cdot x^8 \cdot y^4 \cdot y^1} \\ 11y^2x^4y^2\sqrt{y} \\ = \boxed{11x^4y^4\sqrt{y}} \end{array}$

6-2 Learning Goal: Multiply, divide and simplify radical expressions.

Express in simplest radical form.

10. $\sqrt{30} \cdot \sqrt{6}$

$$= \sqrt{180}$$

$$\begin{array}{c} 2 \cdot 90 \\ 2 \cdot 45 \\ 3 \cdot 15 \\ 3 \cdot 5 \end{array}$$

$$= \sqrt{2 \cdot 2 \cdot 3 \cdot 3 \cdot 5}$$

$$= \boxed{6\sqrt{5}}$$

11. $\sqrt[3]{4x} \cdot \sqrt{7x}$

indexes do not match
so can't multiply
and nothing simplifies

$$= \boxed{\sqrt[3]{4x} \cdot \sqrt{7x}}$$

12. $\sqrt{9x^3y} \cdot \sqrt{3y^4}$

$$\sqrt{27x^3y^5}$$

$$\sqrt{3 \cdot 3 \cdot 3 \cdot x^2 \cdot x \cdot y^4 \cdot y}$$

$$= \boxed{3xy^2\sqrt{3xy}}$$

13. $(\sqrt[3]{8x^5})(\sqrt[3]{2x^2})$

$$= \sqrt[3]{16x^7}$$

$$= \sqrt[3]{2 \cdot 2 \cdot 2 \cdot 2 \cdot x^6 \cdot x}$$

$$= \boxed{2x^2\sqrt[3]{2x}}$$

14. $\frac{\sqrt{80}}{8}$

$$\sqrt{80}$$

$$\begin{array}{c} 2 \cdot 40 \\ 2 \cdot 20 \\ 2 \cdot 10 \\ 2 \cdot 5 \end{array}$$

$$= \frac{\sqrt{2 \cdot 2 \cdot 2 \cdot 2 \cdot 5}}{8}$$

$$= \frac{4\sqrt{5}}{8} = \boxed{\frac{\sqrt{5}}{2}}$$

15. $\frac{\sqrt{28b^{112}}}{\sqrt{b^3}}$ Subtract exponents

$$= \sqrt{28b^{109}}$$

$$= \sqrt{2 \cdot 2 \cdot 7 \cdot b^{108} \cdot b}$$

$$= \boxed{2b^{54}\sqrt{7b}}$$

16. $\frac{\sqrt{224x^2}}{\sqrt{8x^2}}$

$$= \sqrt{\frac{224x^2}{8x^2}}$$

$$= \sqrt{28}$$

$$= \sqrt{2 \cdot 2 \cdot 7}$$

$$= \boxed{2\sqrt{7}}$$

17. $\frac{2}{\sqrt{40}} \cdot \frac{\sqrt{40}}{\sqrt{40}}$

$$= \frac{2\sqrt{40}}{40} = \frac{\sqrt{40}}{20}$$

$$= \frac{\sqrt{2 \cdot 2 \cdot 2 \cdot 5}}{20}$$

$$= \frac{2\sqrt{10}}{20} = \boxed{\frac{\sqrt{10}}{10}}$$

18. $\frac{3y}{6\sqrt{x}} \cdot \frac{\sqrt{x}}{\sqrt{x}}$

$$= \frac{3y\sqrt{x}}{6x}$$

$$= \boxed{\frac{y\sqrt{x}}{2x}}$$

6-3 Learning Goal: I can add, subtract and simplify radical expressions. I can multiply and divide binomial radical expressions

Simplify each of the following and write in simplified radical form:

19. $2\sqrt[3]{2} - 3\sqrt[3]{2} + 4\sqrt{5} + 2\sqrt{5}$ $- \sqrt[3]{2} + 6\sqrt{5}$	20. $6\sqrt{6} - 9\sqrt{6}$ $- 3\sqrt{6}$	21. $-5\sqrt{162} - 5\sqrt{32}$ $- 5\sqrt{2 \cdot 3 \cdot 3 \cdot 3 \cdot 3} - 5\sqrt{2 \cdot 2 \cdot 2 \cdot 2 \cdot 2}$ $- 45\sqrt{2} - 20\sqrt{2}$ $- 65\sqrt{2}$
22. $(3 - \sqrt{5})(3 + \sqrt{5})$ FOIL $9 + 3\sqrt{5} - 3\sqrt{5} - \sqrt{25}$ $9 - 5$ 4	23. $(-3 - 3\sqrt{6})(5 + 2\sqrt{6})$ FOIL $-15 - 6\sqrt{6} - 15\sqrt{6} - 6(6)$ $-15 - 21\sqrt{6} - 36$ $- 51 - 21\sqrt{6}$	24. $\frac{6}{3 + \sqrt{15}}$ $\frac{6}{3 + \sqrt{15}} \cdot \frac{3 - \sqrt{15}}{3 - \sqrt{15}}$ $= \frac{18 - 6\sqrt{15}}{3 - 3\sqrt{15} + 3\sqrt{15} - (\sqrt{15})(\sqrt{15})}$ $= \frac{18 - 6\sqrt{15}}{3 - 15} = \frac{18 - 6\sqrt{15}}{-6}$ $= -3 + \sqrt{15}$
25. $\frac{3 + \sqrt{2}}{\sqrt{5} - \sqrt{10}}$ $\frac{3 + \sqrt{2}}{\sqrt{5} - \sqrt{10}} \cdot \frac{(\sqrt{5} + \sqrt{10})}{(\sqrt{5} + \sqrt{10})}$ $\frac{3\sqrt{5} + 3\sqrt{10} + \sqrt{10} + \sqrt{20}}{5 + \sqrt{50} - \sqrt{50} - 10}$ $= \frac{3\sqrt{5} + 4\sqrt{10} + \sqrt{2 \cdot 2 \cdot 5}}{5 - 10}$ $= \frac{3\sqrt{5} + 4\sqrt{10} + 2\sqrt{5}}{-5}$ $= \frac{5\sqrt{5} + 4\sqrt{10}}{-5}$		

6-4 Learning Goal: I can simplify expressions with rational exponents.

Simplify each of the following and write in rational exponent form.

26. Rewrite the expression in radical form $x^{\frac{1}{3}}$

$$\sqrt[3]{x}$$

27. Rewrite the expression in exponential form. $\sqrt[5]{x^7}$

$$x^{\frac{7}{5}}$$

Express in simplest form

28. $27^{2/3}$

$$\begin{aligned} &(\sqrt[3]{27})^2 \\ &(3)^2 \\ &9 \end{aligned}$$

29. $(125x^6)^{1/3}$

$$\begin{aligned} &\sqrt[3]{125x^6} \\ &\sqrt[3]{5 \cdot 5 \cdot 5 \cdot x^6} \\ &5x^2 \end{aligned}$$

30. $62^{\frac{3}{2}}$

$$\begin{aligned} &\sqrt[2]{62^3} \\ &= \sqrt{62 \cdot 62 \cdot 62} \\ &62\sqrt{62} \end{aligned}$$